Divisibility
and
Modular Arithmetic

DIVISION

Definition:

If a and b are integers with $a \neq 0$, we say that a *divides* b if there is an integer c such that $b = ac$, or equivalently, if b/a is an integer. When a divides b we say that a is a *factor* or *divisor* of b, and that b is a *multiple* of a. The notation $a \mid b$ denotes that a divides b. We write $a \nmid b$ when a does not divide b.

DIVISION

Example 1:

Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Solution:

We see that $3 \nmid 7$, because $7/3$ is not an integer. On the other hand, $3 \mid 12$ because $12 / 3 = 4$.

DIVISION

Theorem:

Let a, b, and c be integers, where $a \neq 0$. Then

(i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

(ii) if $a \mid b$, then $a \mid bc$ for all integers c;

(iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Direct Proof of (i)

Suppose that $a \mid b$ and $a \mid c$. Then, from the definition of divisibility, it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t).$$

DIVISION

Theorem: Division Algorithm

Let a be an integer and d a positive integer. Then there are unique integers q and r, with $0 \leq r < d$, such that $a = dq + r$.

Definition:

In the equality given in the division algorithm, d is called the divisor, a is called the dividend, q is called the quotient, and r is called the remainder. This notation is used to express the quotient and remainder:

$$q = a \text{ div } d, \; r = a \bmod d.$$

DIVISION

Example 2:

What are the quotient and remainder when 101 is divided by 11?

Solution:

We have

$$101 = 11 \cdot 9 + 2.$$

Hence, the quotient when 101 is divided by 11 is 9 = 101 **div** 11, and the remainder is 2 = 101 **mod** 11.

DIVISION

Example 3:

What are the quotient and remainder when -11 is divided by 3?

Solution:

$$-11 = 3(-4) + 1$$

Hence, the quotient when -11 is divided by 3 is $-4 = -11$ **div** 3, and the remainder is $1 = -11$ **mod** 3.

Note that the remainder cannot be negative. Consequently, the remainder is not -2, even though

$$-11 = 3(-3) - 2,$$

because $r = -2$ does not satisfy $0 \leq r < 3$.

MODULAR ARITHMETIC

Definition:

If a and b are integers and m is a positive integer, then a is *congruent to b modulo m* if m divides $a - b$. We use the notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m. We say that $a \equiv b \pmod{m}$ is a **congruence** and that m is its **modulus** (plural **moduli**). If a and b are not congruent modulo m, we write $a \not\equiv b \pmod{m}$.

Note:

$a \equiv b \pmod{m}$ - represents a relation on the set of integers

a **mod** $m = b$ - represents a function

MODULAR ARITHMETIC

Theorem:

Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if a **mod** $m = b$ **mod** m.

Example:

Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6

Solution:

Because 6 divides $17 - 5 = 12$, we see that $17 \equiv 5 \pmod{6}$. However, because $24 - 14 = 10$ is not divisible by 6, we see that $24 \not\equiv 14 \pmod{6}$.

MODULAR ARITHMETIC

Theorem:

Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Theorem:

Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

MODULAR ARITHMETIC

Example:

Because $7 \equiv 2 \pmod 5$ and $11 \equiv 1 \pmod 5$, it that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod 5$$

and that

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod 5.$$

Primes
and
Greatest Common Divisors

PRIME

- An integer p greater than 1 is called ***prime*** if the only positive factors of p are 1 and p.
- A positive integer that is greater than 1 and is not prime is called ***composite***.
- **Example**:

  The integer 7 is prime because its only positive factors are 1 and 7, whereas the integer 9 is composite because it is divisible by 3

PRIME

- Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.
- **Example**: The prime factorizations of 100, 641, 999, and 1024 are given by

  $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2252,$

  $641 = 641,$

  $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37,$

  $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

PRIME

- **Mersenne prime -** is a prime number of the form $M_n = 2^n - 1$ for some integer n.
- **Example**:

  $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$ and $2^7 - 1 = 127$ are Mersenne prime

  While,

  $2^{11} - 1 = 2047$ is not a Mersenne prime because $2047 = 23 \cdot 89$

PRIME

- **Twin primes** are pairs of primes that differ by 2
- **Example**:

  3 and 5
  5 and 7
  11 and 13
  17 and 19
  4967 and 4969

GREATEST COMMON DIVISORS

- Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of a and b.
- The greatest common divisor of a and b is denoted by gcd(a, b).
- Example:
  1. What is the greatest common divisor of 24 and 36?

     Solution: The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, gcd(24, 36) = 12.

  2. What is the greatest common divisor of 17 and 22?

     Solution: The integers 17 and 22 have no positive common divisors other than 1, so that gcd(17, 22) = 1.

GREATEST COMMON DIVISORS

- The integers a and b are **relatively prime** if their greatest common divisor is 1.
- Example: Determine whether the following integers are pairwise relatively prime.
  - 10, 17 and 21

    Solution: pairwise relatively prime bec. gcd(10, 17) = 1, gcd(10, 21) = 1, and gcd(17, 21) = 1

  - 10, 19 and 24

    Solution: not pairwise relatively prime bec. gcd(10, 24) = 2 > 1

GREATEST COMMON DIVISORS

- Prime factorizations can be used to find the greatest common divisor of two positive integers
- Suppose that the prime factorizations of the positive integers *a* and *b* are

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \; b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n},$$

where each exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either *a* or *b* are included in both factorizations, with zero exponents if necessarv. Then *acd(a*. *b*) is given bv

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)},$$

- Example: the prime factorizations of 120 and 500 are $120 = 2^3 \cdot 3 \cdot 5$ and $500 = 2^2 \cdot 5^3$, the greatest common divisor is

$$\gcd(120, 500) = 2^{\min(3, 2)} 3^{\min(1, 0)} 5^{\min(1, 3)} = 2^2 3^0 5^1 = 20.$$

LEAST COMMON MULTIPLE

- The **least common multiple** of the positive integers *a* and *b* is the smallest positive integer that is divisible by both *a* and *b*. The least common multiple of *a* and *b* is denoted by lcm*(a, b)*.
- Prime factorizations can also be used to find the least common multiple of two integers.
- Then the least common multiple of *a* and *b* is given by

$$\operatorname{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)},$$

- Example: What is the least common multiple of 233572 and 2433?

Solution:

$$\operatorname{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^{\max(3, 4)} 3^{\max(5, 3)} 7^{\max(2, 0)} = 2^4 3^5 7^2.$$

EUCLIDEAN ALGORITHM

- more efficient method of finding the greatest common divisor
- Let $a = bq + r$, where a, b, q, and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.
- Example:

  Finding the GCD of 414 and 662 using Euclidean algorithm

  Solution: Successive uses of the division algorithm give:

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41$$

Hence, $\gcd(414, 662) = 2$, because 2 is the last nonzero remainder.

GCDs as LINEAR COMBINATIONS

- The GCD of two integers a and b can be expressed in the form

  $$sa + tb$$

  where s and t are integers. In other words, $gcd(a, b)$ can be expressed as a **linear combination** with integer coefficients of a and b.
- Bezout's Theorem

  If a and b are positive integers, then there exist integers s and t such that

  $$gcd(a,b) = sa + tb$$
- Example: $\gcd(6, 14) = 2$, and $2 = (-2) \cdot 6 + 1 \cdot 14$

GCDs as LINEAR COMBINATIONS

Definition :

If a and b are positive integers, then integers s and t such that $gcd(a, b) = sa + tb$ are called Bézout coefficients of a and b (after Étienne Bézout, a French mathematician of the eighteenth century). Also, the equation $gcd(a, b) = sa + tb$ is called Bézout's identity.